

ACG Architecture & System Design

Merchant-Side Control Plane for AI-Originated Transactions

1. System Pipeline & Core Boundaries

The Agent Commerce Gateway operates as an explicit, 6-phase zero-trust pipeline. No financial action can proceed directly from an AI model to a payment rail without passing all authorization gates.



2. Separation of Concerns

- AI Proposer (Zero Authority):** AI agents, LLMs, and subagents synthesize buyer intent and propose items, quantities, and client nonces. They possess **zero authority** to set prices, modify taxes, or self-authorize execution.
- ACG Control Plane (Deterministic Authorizer):** ACG inspects cryptographic delegation, validates against merchant database truth, applies active merchant policy, and acquires ACID locks across budget and stock.
- Razorpay Rails (Financial Executor):** Razorpay receives pre-authorized, idempotent orders, captures customer funds, executes refunds, and dispatches HMAC-signed webhook notifications.

3. Financial Action Types

ACG models explicit financial action types:

* **AUTHORIZE_AND_RESERVE:** Evaluates intent against policy and locks budget and inventory.

- * **CREATE_PAYMENT_ORDER**: Invokes Razorpay POST /v1/orders with receipt = intent_id.
- * **CAPTURE_PAYMENT**: Processes Razorpay payment.captured webhook and commits dual-resource locks.
- * **RELEASE_RESERVATION**: Aborts and rolls back reserved budget and inventory upon timeout, cancellation, or rail failure.
- * **EXECUTE_REFUND**: Issues idempotent refund via X-Refund-Idempotency when post-capture fulfillment fails.

4. State Transition Machine

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[INTENT_RECEIVED]
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▼
[INTENT_VALIDATED] ==> [INTENT_REJECTED] (If signature/truth/policy fails)
|
▼
[DUAL_RESERVATION_HELD] ==> [RESERVATION_FAILED] (If stock/budget exhausted)
|
▼
[ORDER_CREATED] ==> [DUAL_RESERVATION_RELEASED] (If Razorpay order API fails)
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▼ ▼
[PAYMENT_CAPTURED] [PAYMENT_FAILED]
| |
▼ ▼
[FULFILLMENT_DISPATCHED] [DUAL_RESERVATION_RELEASED]
|
▼ (If warehouse failure)
[REFUND_PENDING]
|
▼
[REFUNDED]
```